

SatQuery-Edge

Offline Satellite Intelligence Investigation Report
SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-20260902-P2M9GZ
Timestamp	2026-09-02 18:56:39
Intent	text_guided_grounding
Tools Used	grounding, geospatial

Query

Detect forest fire damage and highlight burn scar vegetation loss regions in Uttarakhand.

Final Answer

Investigation complete. Evidence Score: 83/100.

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.901	0.30
Sensor Reliability	S	0.860	0.20
Spatial Consistency	P	0.868	0.20
Temporal Consistency	T	0.820	0.15
Input Quality	I	0.609	0.15
FINAL SCORE		83.0 / 100	

Validation

Status: **PASS** ✓

- 1 image(s) detected — count OK.
- Image 1: Quality OK (mean=100, std=42.7).

Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km²)	Demo Georeferenced
1	candidate vegetation	0.82	157.205	Yes

Specialist Tool Outputs

GROUNDING — SUCCESS — 1 region(s)

Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'vegetation'.
Highest confidence: 0.82.

GEOSPATIAL — SUCCESS — 0 region(s)

Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 157.21 km².

■ Approximate demo georeferencing applied.
Demo center: (20.296°N, 85.824°E)
Coordinates are NOT derived from real GeoTIFF metadata.

Execution Trace

[18:56:39] QUERY RECEIVED: Detect forest fire damage and highlight burn scar vegetation loss regions in Uttarakhand.
[18:56:39] ROUTER DECISION: Intent: text_guided_grounding
Selected tools: grounding, geospatial
Images required: 1
[18:56:39] VALIDATION: Status: PASS
1 image(s) detected – count OK.
Image 1: Quality OK (mean=100, std=42.7).
[18:56:39] TOOL EXECUTING: grounding → started
[18:56:39] TOOL OUTPUT (grounding): Success – 1 region(s) detected
Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'vegetation'.
Highest confidence: 0.82.
[18:56:39] TOOL EXECUTING: geospatial → started
[18:56:39] TOOL OUTPUT (geospatial): Success – 0 region(s) detected
Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 157.21 km².
[18:56:39] EVIDENCE FUSION: Score: 83.0/100
M=0.901
S=0.860
P=0.868
[18:56:39] FINAL SCORE: 83/100
[18:56:39] ANSWER: Investigation complete. Evidence Score: 83/100.

Tool Versions

Component	Version
SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0
Report Engine	ReportLab PDF
Mode	OFFLINE — No external APIs

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge